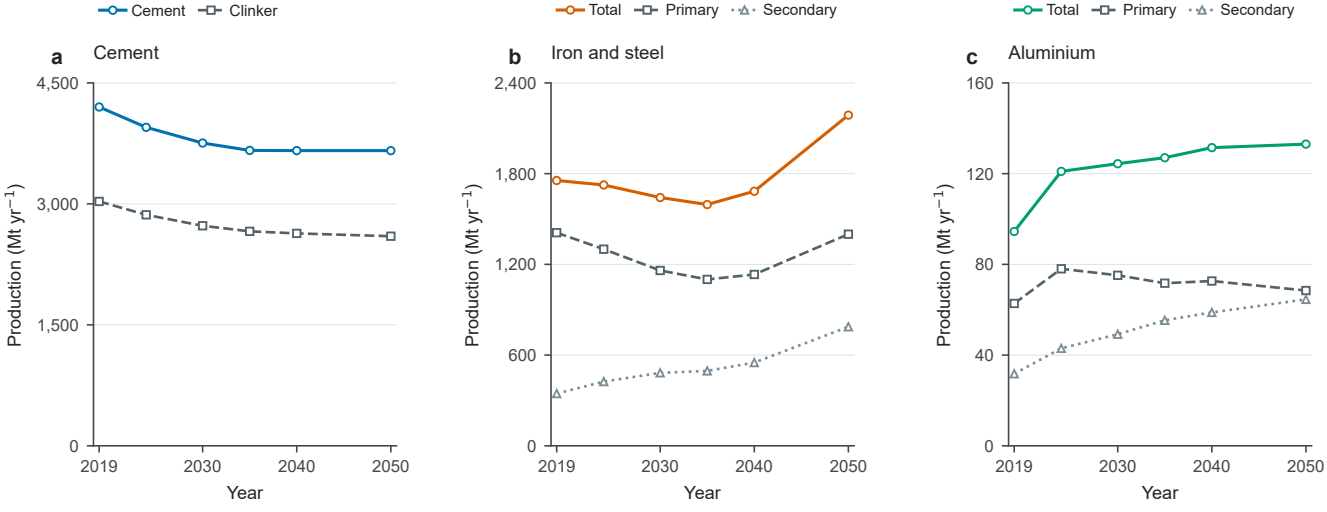
Global material production and production routes

No circular economy | 2019–2050

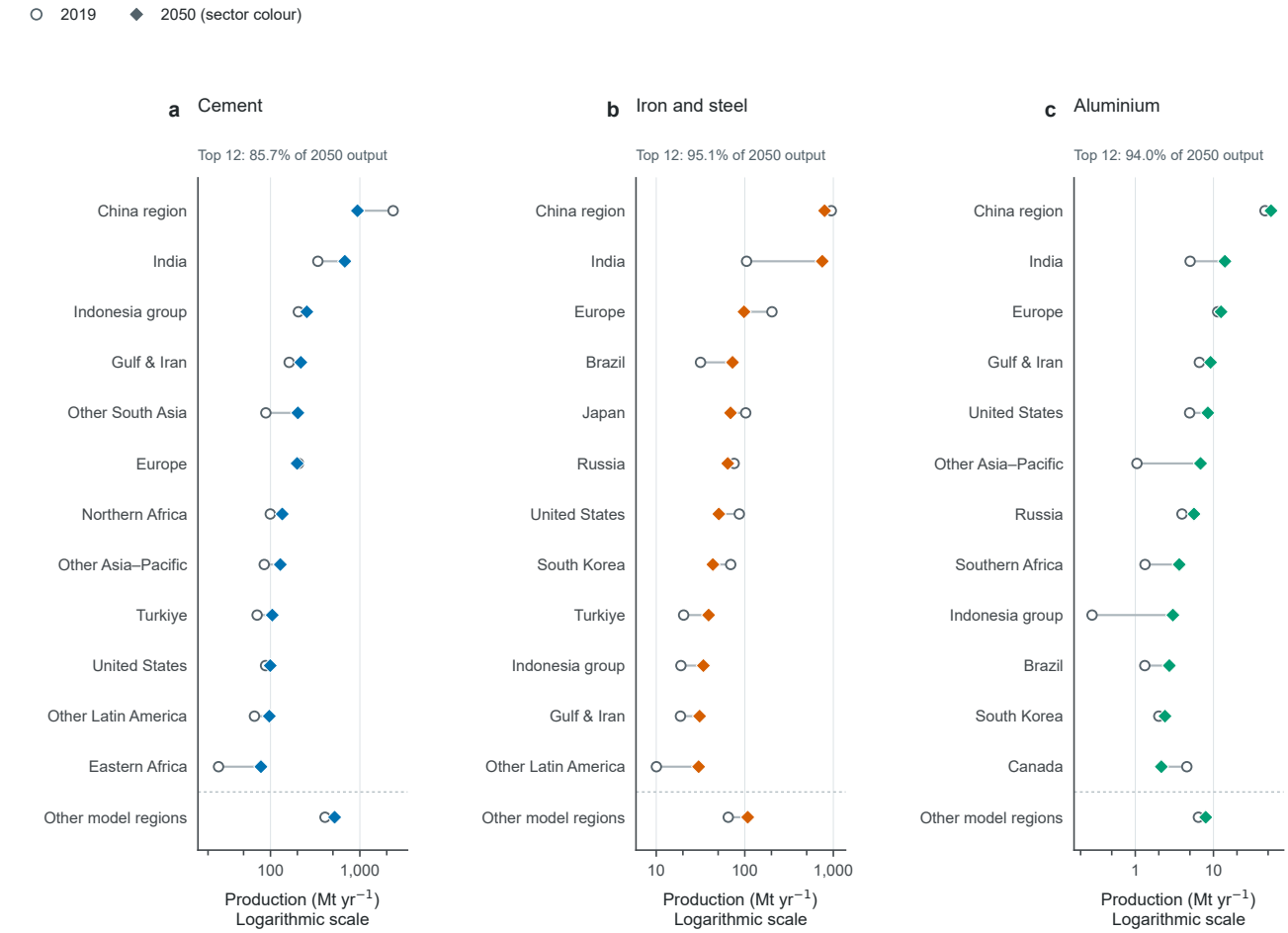


Metals: total = primary + secondary. Cement and clinker are distinct products.

Draft assumption: missing additive activity = zero.

The geography of industrial production

No circular economy | 2019–2050



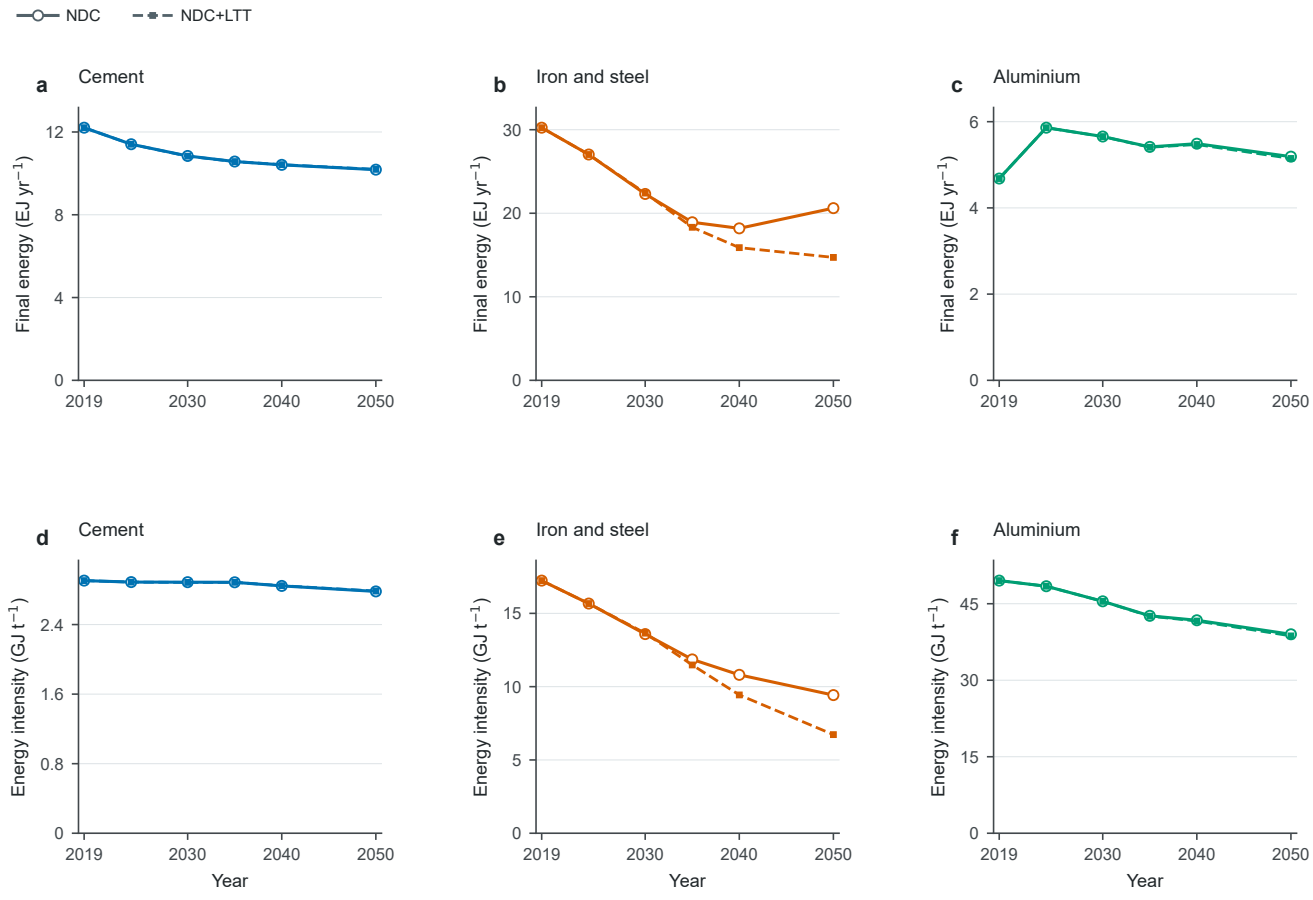
Turkiye denotes MDA: Turkiye, Israel, Syria, Jordan, Lebanon and Palestine.

Europe combines five model regions. Indonesia group = Indonesia, Philippines and Viet Nam.

Draft assumption: missing additive activity = zero.

Industrial energy requirements

No circular economy | 2019–2050

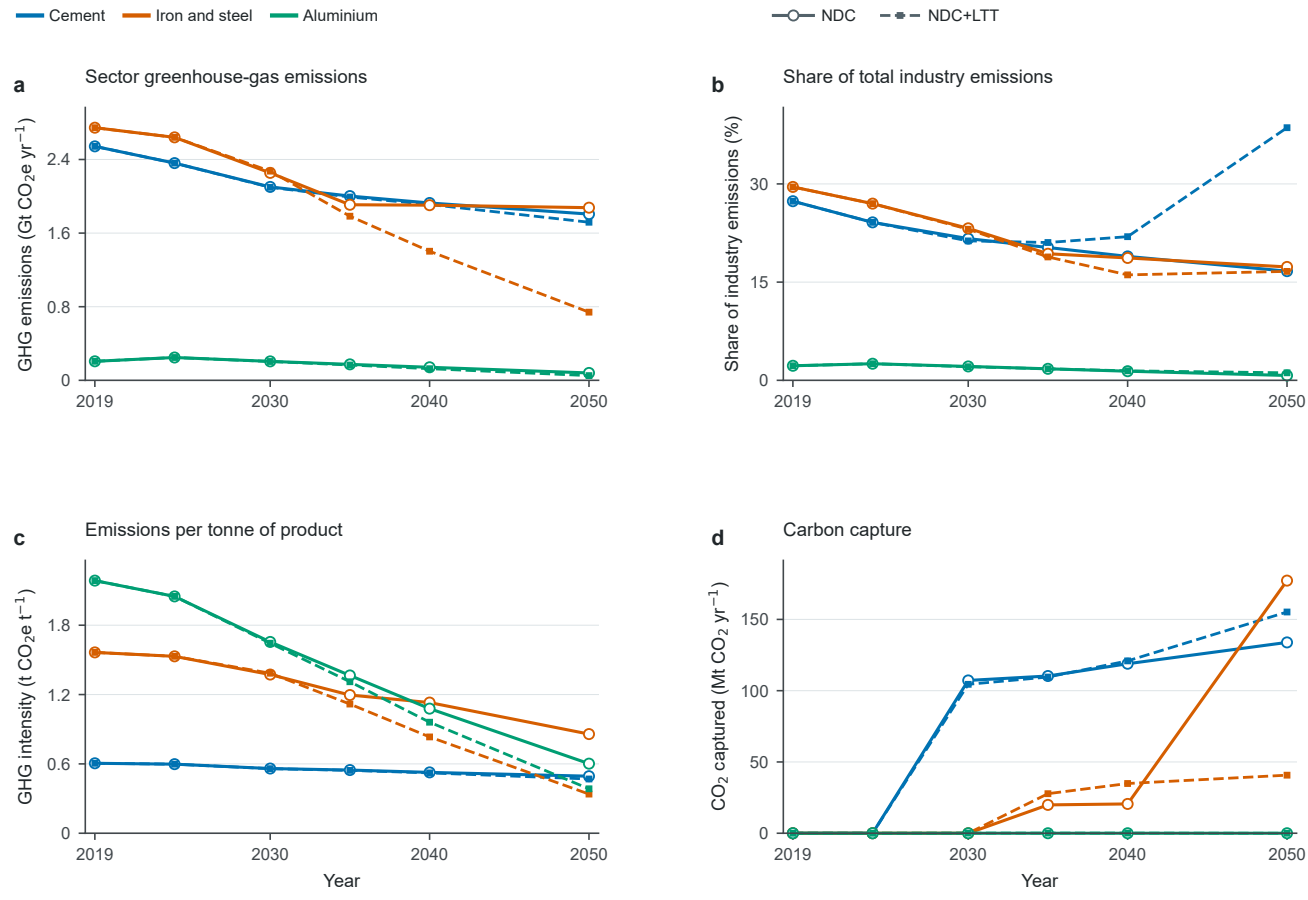


Energy intensity is the ratio of global final energy to global material production.

Draft assumption: missing additive activity = zero.

Industrial emissions and carbon capture

No circular economy | 2019–2050

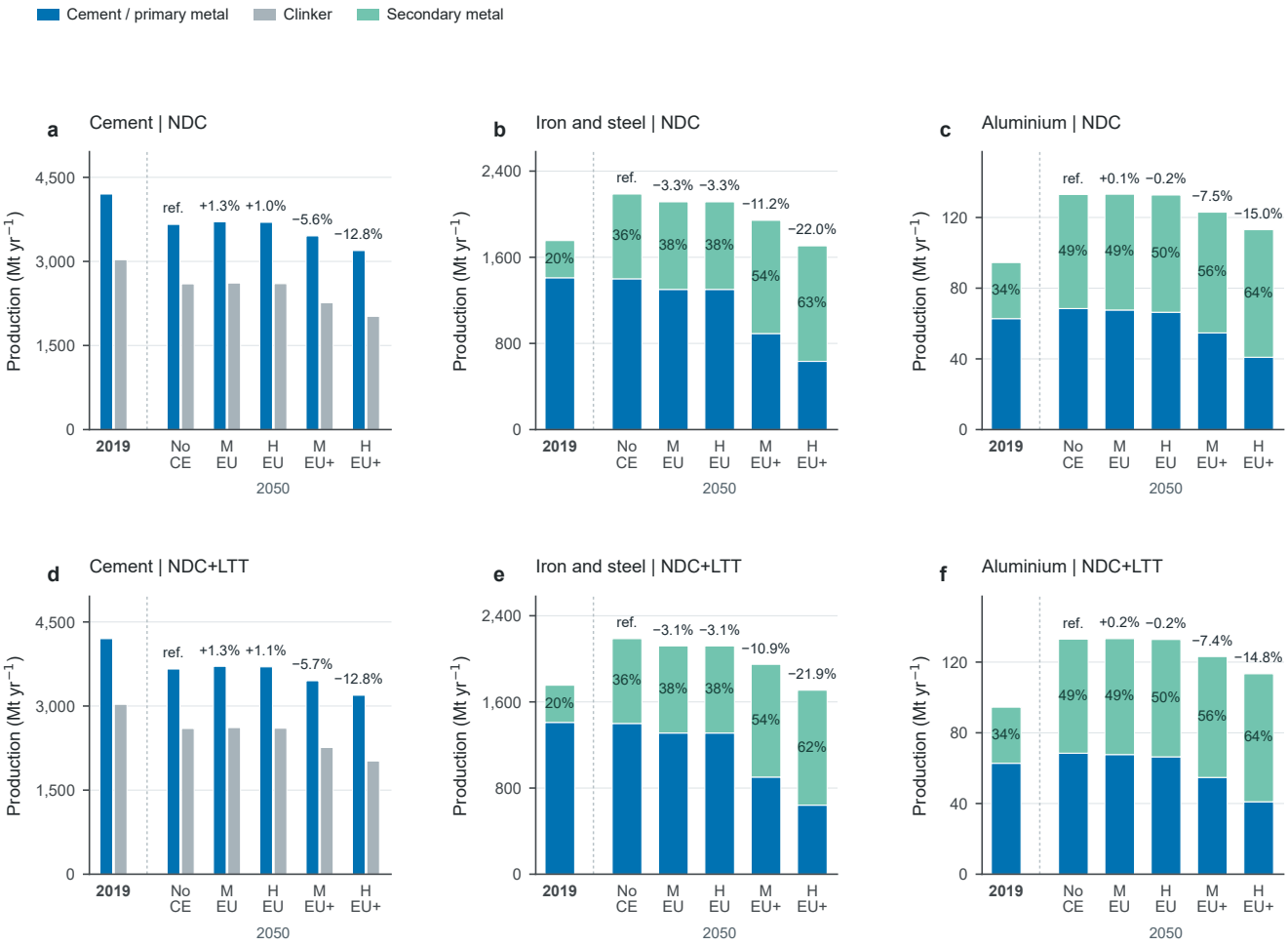


GHG emissions are the reported model values. CO2 capture is shown separately.

Draft assumption: missing additive activity = zero.

Circular economy changes material demand and primary supply

Global production | 2019 reference and 2050 CE scenarios



2019: no CE. All other bars: 2050. Labels above: change from the same pathway's 2050 no CE.

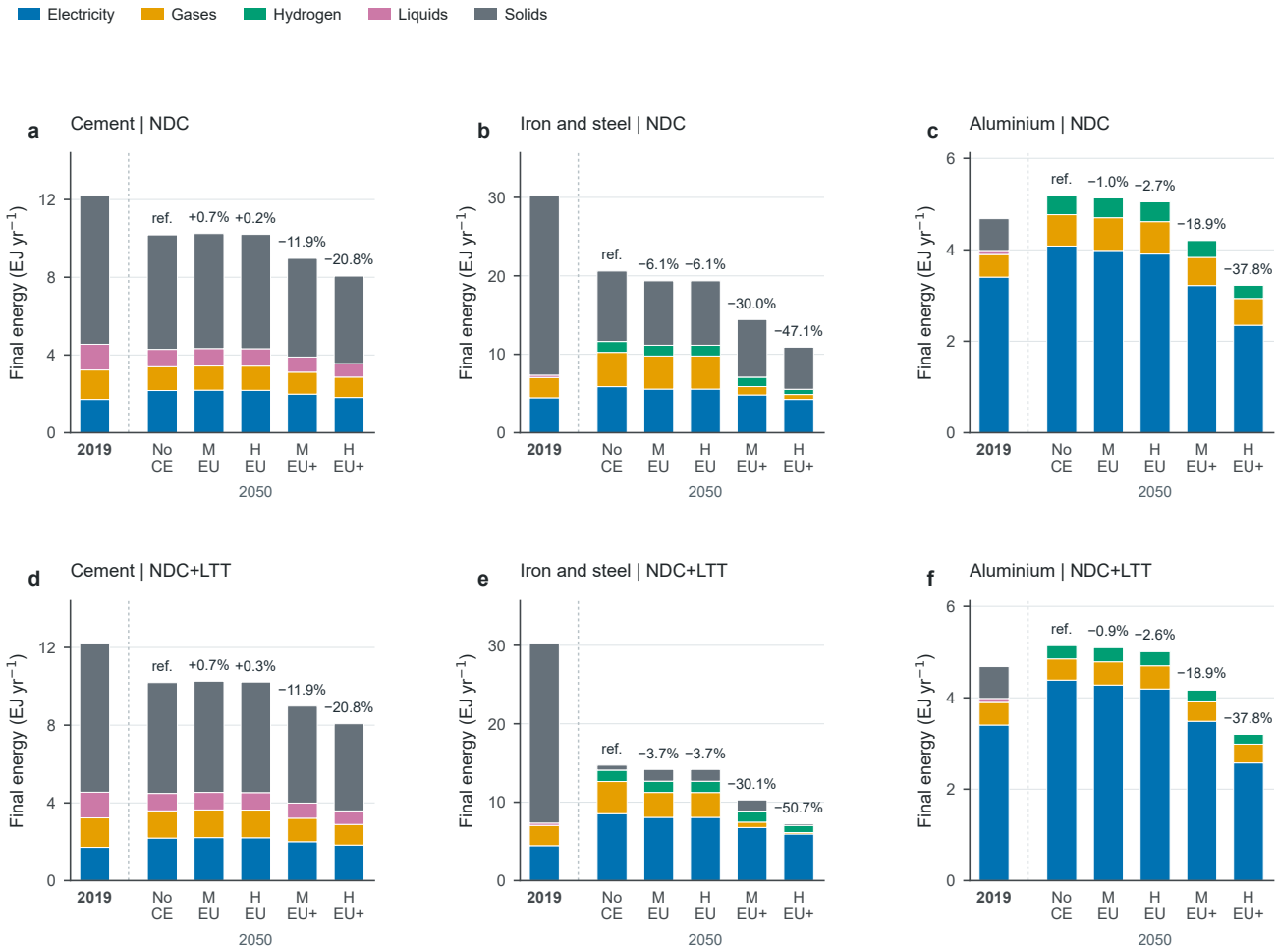
Changes refer to cement or total metal output. Within metal stacks: secondary-production shares.

M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Draft assumption: missing additive activity = zero.

Circular economy reduces final energy demand and changes the fuel mix

Global final energy | 2019 reference and 2050 CE scenarios



2019: no CE. All other bars: 2050. Labels above: change from the same pathway's 2050 no CE.

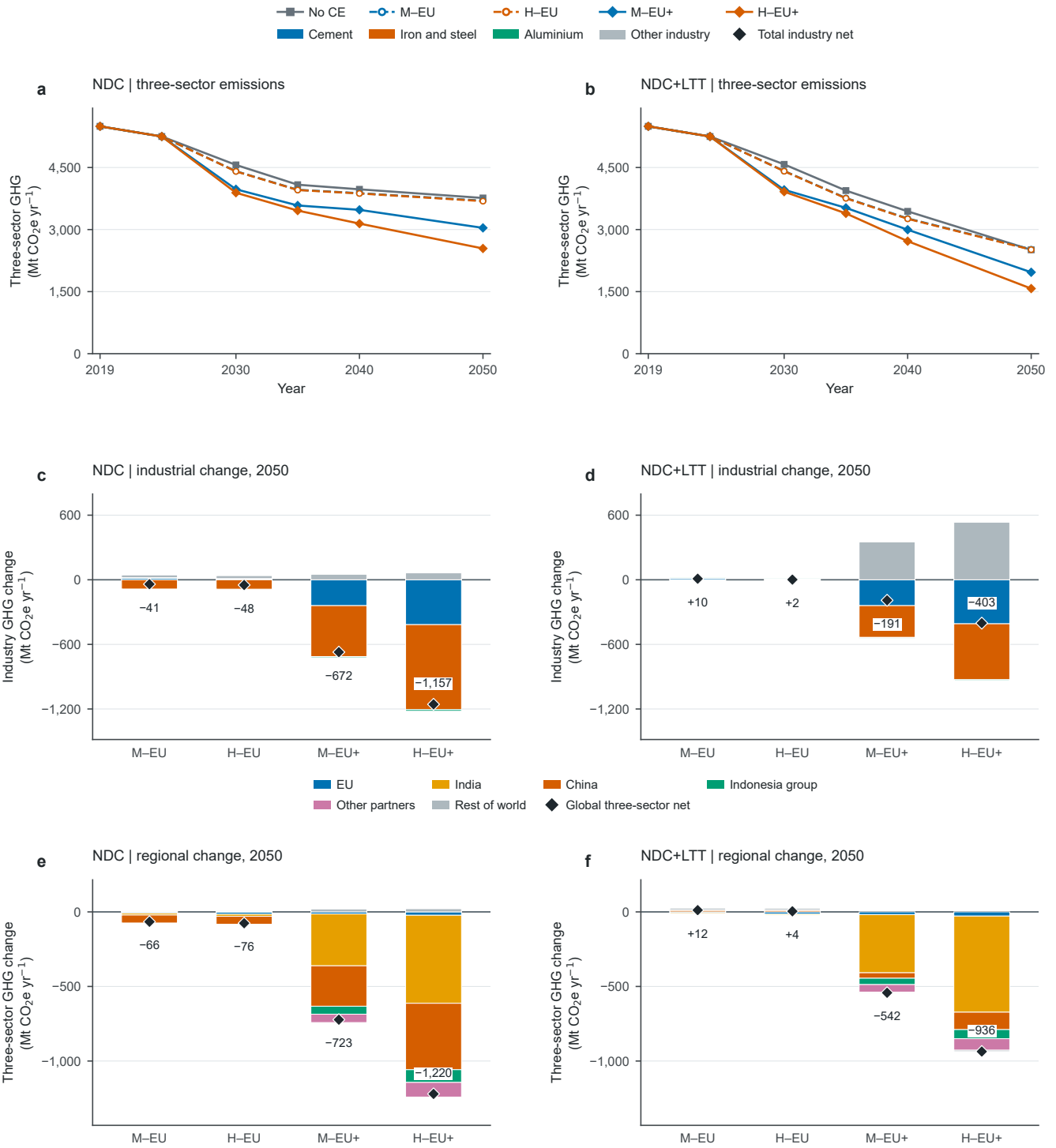
Each sector uses the same scale across both climate pathways; sectors have different scales.

M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Draft assumption: missing additive activity = zero.

CE emissions savings, industrial responses and international cooperation

Global GHG trajectories, 2019–2050 | matched 2050 changes from no CE



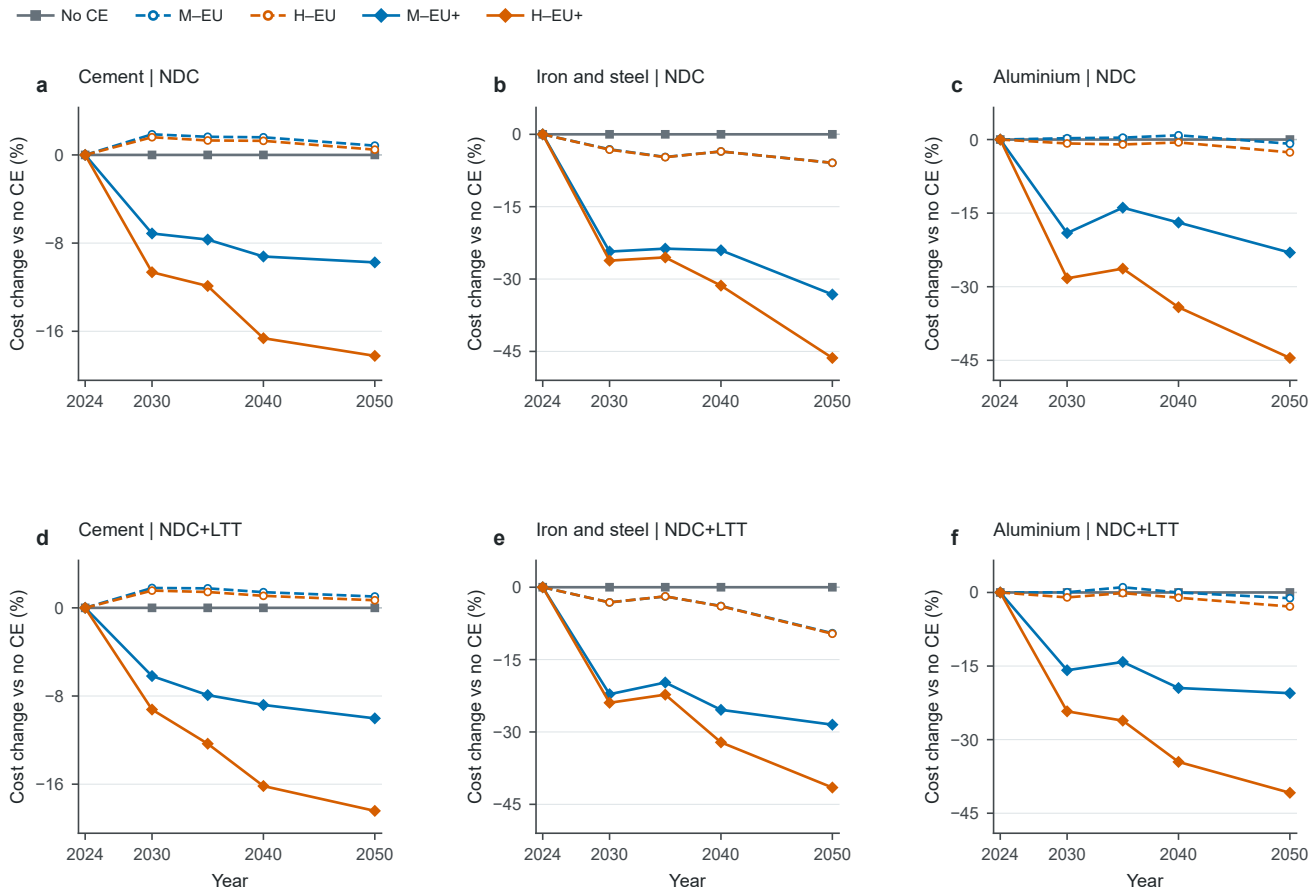
Diamonds: middle = all-industry net change; bottom = global three-sector net change.

M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Draft assumption: missing additive activity = zero.

Circular economy changes annualised industrial sector costs

Change from the same year's no-CE cost | 2024–2050



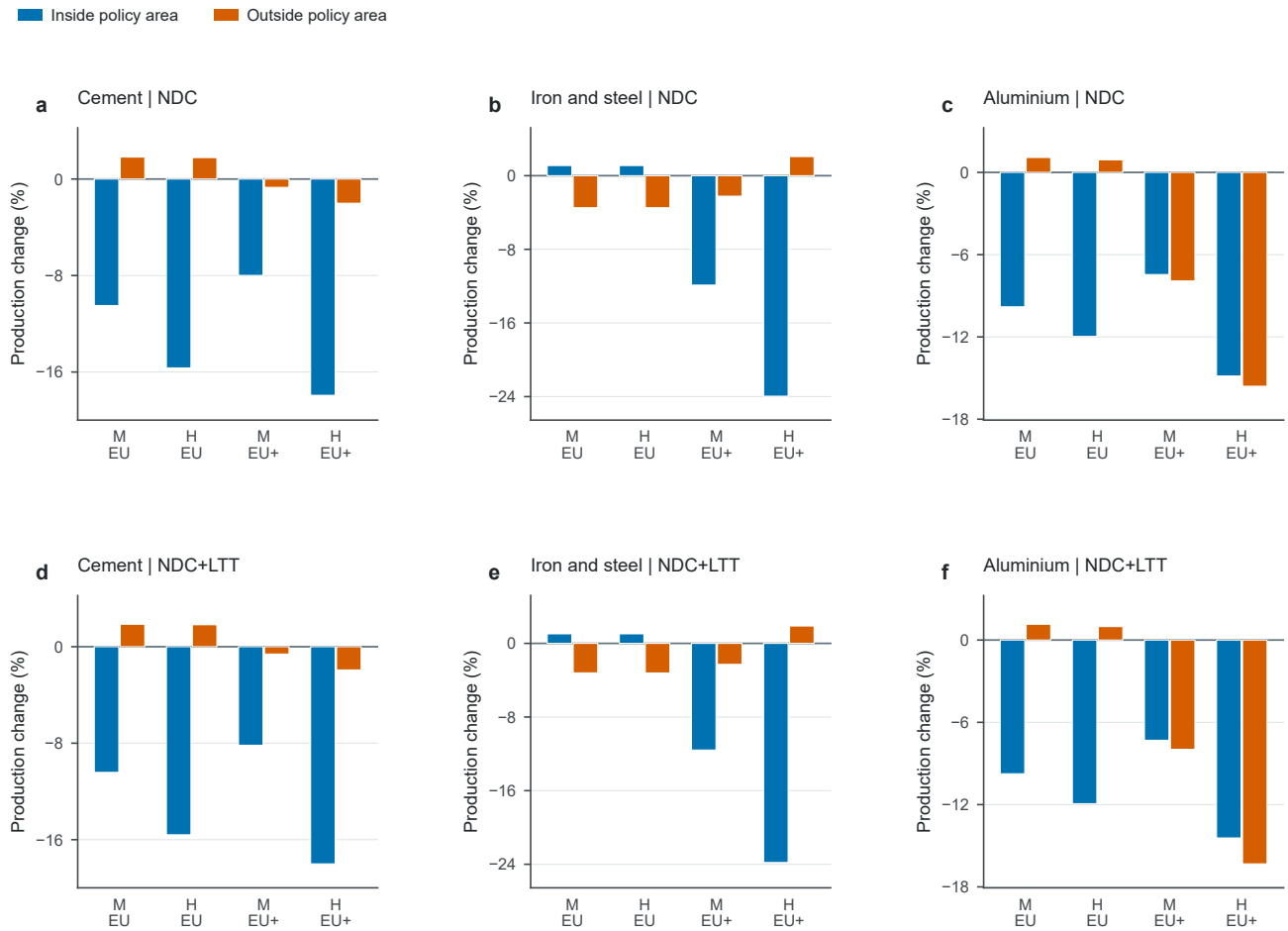
M/H = medium/high CE; EU+ = EU + international partners (17 model regions).

Reference: same sector, climate pathway and year. Source 2019 costs are retained in the data.

Costs retain missingness; unavailable or nonpositive no-CE costs make percentage changes undefined.

Material production inside and outside CE policy areas

Percentage change from matched no CE | 2050



M/H = medium/high CE; EU+ = EU + international partners. Zero = matched no CE.

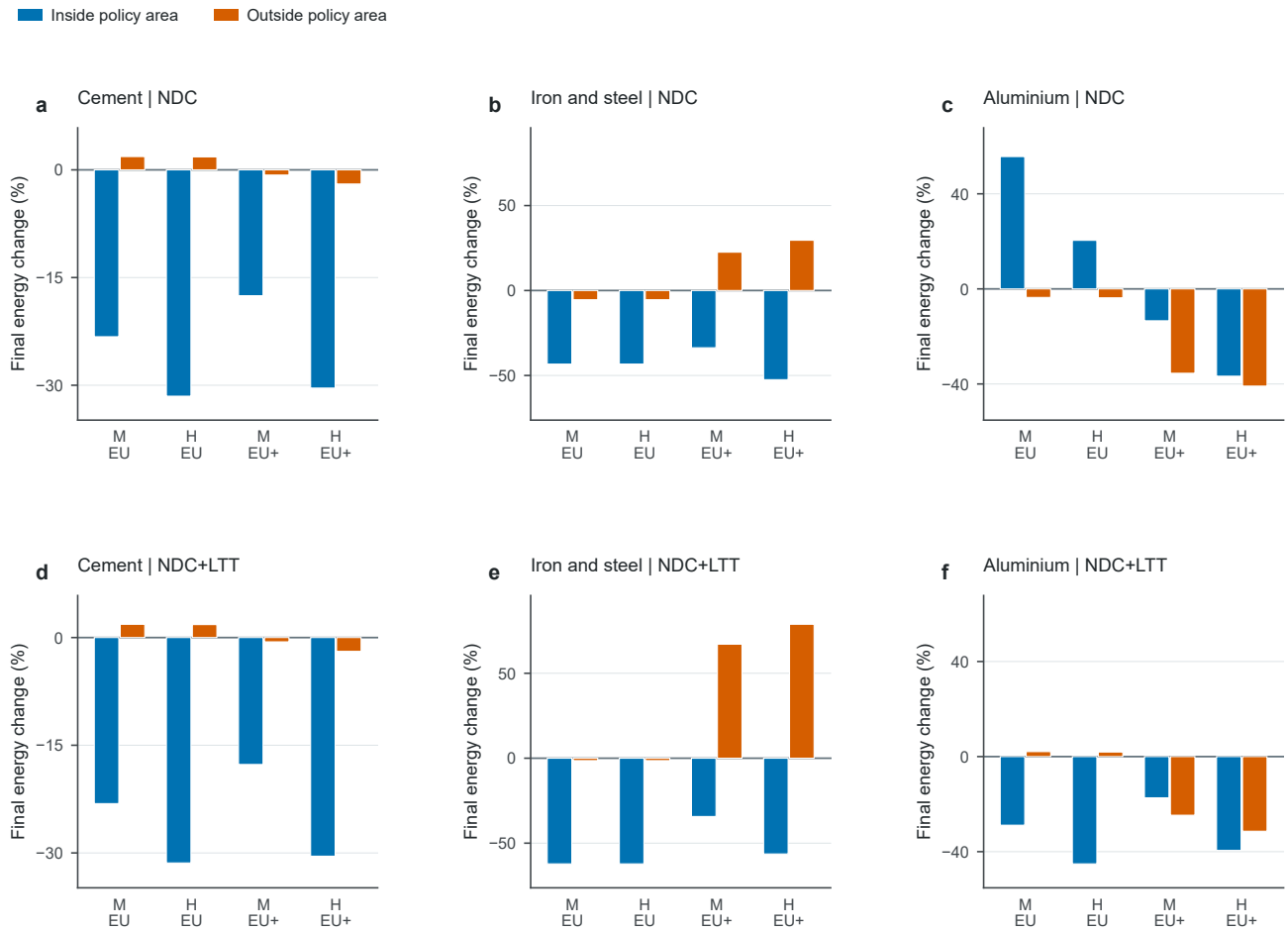
EU: 3 inside / 25 outside; EU+: 17 inside / 11 outside model regions.

Production = cement output or primary + secondary metal output.

Draft assumption: missing additive activity = zero. Undefined percentages appear as gaps.

Final energy demand inside and outside CE policy areas

Percentage change from matched no CE | 2050



M/H = medium/high CE; EU+ = EU + international partners. Zero = matched no CE.

EU: 3 inside / 25 outside; EU+: 17 inside / 11 outside model regions.

Final energy sums exact parent carriers, excluding nested fuel subcategories.

Draft assumption: missing additive activity = zero. Undefined percentages appear as gaps.